



Grassland restorations improve pollinator communities: a meta-analysis

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Abstract

Natural grasslands are being destroyed at an alarming pace, but land managers are actively working to restore these habitats. Many of these efforts focus on restoring plant diversity but often do not consider responses of higher trophic levels such as pollinators, which provide crucial ecosystem services. We conducted a meta-analysis of 25 large-scale studies to examine the effects of grassland restorations on pollinator communities. Specifically, we compared pollinator communities in restored, degraded and remnant grasslands to determine if restorations improve pollinators from a degraded state and if they fully restore them to remnant values. We found that grassland restorations significantly improved both pollinator abundance and richness as compared to degraded grasslands. Additionally, pollinator abundance and richness did not significantly differ from remnant sites, indicating a near full recovery. Sub-analyses found that factors such as pollinator taxa, restoration age, and mode of land degradation all influenced the magnitude of recovery. In particular, lepidopteran abundance increased more than bee abundance in these restorations. Older restorations (> 10 years) showed the strongest improvements in pollinator communities. This research highlights the importance of grassland restorations in supporting not only plant diversity but also pollinators.

Keywords Degraded grassland · Conservation management · Bees · Lepidoptera · Remnant grassland

Introduction

Native prairie and grassland habitats provide a wide diversity of ecosystem services, including the support of pollinator communities due to their high forb richness and abundance (Kearns et al. 1998; Ghazoul 2006). However, native grasslands have suffered widespread habitat destruction, often via conversion to agriculture, and in some regions have been reduced to just 0.1% of their historical cover (Samson and Knopf 1994; Moranz et al. 2012). Aside from managed species such as the European honeybee (*Apis mellifera*), native and wild pollinators cannot flourish without a diverse floral community both spatially and temporally (Ogilvie and

Forrest 2017), and so the loss of native grasslands has been detrimental to pollinators as it has homogenized the landscape and reduced foraging and nesting resources. This landscape homogenization has especially negative consequences on rare species, dispersal limited species, and specialists (Baur 2014; Borschig et al. 2013). Grassland restorations may enhance pollinator communities along with other ecosystem services, though few studies have focused on pollinator responses to large-scale grassland restoration efforts.

Historically, most land managers have focused on enhancing plant community diversity when implementing grassland restorations (Young 2000). This “plant-first” focus hopes to impact other trophic levels via food web interactions. Cascading effects from the plant community to higher trophic levels have been widely documented (Longcore 2003; Nemec and Bragg 2008; Molano-Flores 2009; Kaiser-Bunbury et al. 2017). For example, in studies by Nemec and Bragg (2008) and Molano-Flores (2009), herbivorous insects responded positively to grassland restoration, showing increased levels of seed herbivory and higher diversity levels in restored prairies. However, while grassland restoration efforts typically improve plant communities, it is

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still unclear whether such restorations are as successful in restoring pollinator communities, in part due to an inability to clearly define pollinator community restoration goals. For example, pollinator abundance and richness are two, sometimes contrasting, aspects of pollinator community diversity. Habitat type and restoration efforts may increase abundance of a single pollinator species but have differing effects on overall pollinator richness (Brown 1984). Efforts to increase *Danaus plexippus* (Monarch butterfly) abundance through creation of “Monarch Waystations” are examples of how restoration efforts may target one species and not an entire community. Further, pollinators may have taxon-specific responses to restoration efforts due to differing life histories, dispersal abilities, and nesting habits (Öckinger et al. 2018). For example, bees tend to respond to both local factors (e.g., habitat quality) and landscape factors (e.g., urbanization) while lepidopteran species are often more responsive to local factors (Sjodin et al. 2008; Munsch et al. 2019; Poniatowski et al. 2018; Williams and Kremen 2007; McCune et al. 2020; Carson et al. 2016; Renaud et al. 2016; Torne-Noguera et al. 2014). That being said, landscape level modifications will likely impact any species, especially at the level of the meta-population and so it should not be discounted.

An additional consideration is the difficulty in defining restoration targets for pollinator communities. For plants, restoration targets are often defined based on historical records (Meine 1999). However, historical records are often lacking for pollinator communities, and so restoration targets are based on pollinator diversity in remnant grasslands (Bartomeus et al. 2019). Similar to plant community responses, a restoration may improve pollinator diversity, but not meet restoration targets when compared to remnant habitats. Finally, it is possible that degraded habitats may actually be better for pollinator communities, especially if they contain invasive plant species with abundant floral resources. For example, Emery and Doran (2013) found that an invasive forb, baby’s breath (*Gypsophila paniculata*), significantly increased pollinator abundances in a sand dune grassland, and restoration efforts had little effect on pollinator communities.

Several technical aspects of grassland restoration efforts may further complicate responses of pollinator communities. Factors such as type of land degradation, restoration methods, or age of restoration can influence pollinator responses. For example, land degraded by an invasive plant species or animal grazing may have higher pollinator abundance and richness than one that was used for row-crop agriculture, as many solitary bees nest in the ground and are negatively impacted by tilling (Williams et al. 2010). Similarly, grassland restorations that involve multiple methods such as tilling or burning may negatively affect pollinators by killing larvae or adults. Finally, as with plant communities, it may take years or even decades for pollinators to colonize

restored grassland patches and reach target diversity goals (Emery and Rudgers 2010).

We conducted a meta-analysis to assess the overall impact of grassland restorations on pollinator communities. Specifically, we asked:

- 1) Do grassland restorations improve pollinator communities compared to degraded sites?
- 2) Do these restorations restore the pollinator communities to remnant conditions?

We followed-up with sub-analyses to address specific aspects of pollinator community responses, namely: Do pollinator abundance and richness show differences in response to restoration efforts? Do pollinator taxa differ in their responses to restoration? Does the type of initial land degradation influence pollinator community responses? Do methods associated with restoration negatively affect pollinators? and Does length of time since restoration influence pollinator responses? We focused only on large-scale restoration efforts (> 1 ha), as most pollinators are capable of relatively long-distance dispersal (Hill et al. 2011; Osborne et al. 1999). Additionally, large-scale restorations better reflect goals for land managers.

We hypothesized that pollinator communities would improve in grassland restorations but would fall short of remnant conditions, similar to the responses of many plant communities (Delaney et al. 2015; Copeland et al. 2018). Additionally, we hypothesized that abundance would be more readily improved than richness, and that these responses would differ among taxa based on differences in floral resource preferences and dispersal abilities. We hypothesized that restorations of grasslands degraded by invasive species would improve pollinator communities more than those degraded by agriculture. We also hypothesized that restorations using multiple restoration methods would harm pollinators, by inadvertently killing off larval or adult populations with practices such as fire or tilling. Finally, we expected older restorations to have increased pollinator diversity compared to younger restorations.

Methods

Literature search

To identify relevant studies, we conducted a search on Web of Science on April 12, 2019 using the following key terms “prairie restor* AND (pollinat* abundance OR pollinat* richness OR pollinat* diversity OR bee abundance OR butterfly abundance OR moth abundance)”, “savannah restor* AND pollinat*”, “prairie restor* AND bird pollinat*”, “meadow restor* AND pollinat*”, “grassland restor* and

pollinat*. We also conducted haphazard searches in Google Scholar with similar keywords. In addition, we searched the reference lists of the papers selected to include studies not found by database searches. These searches yielded 416 papers.

Following these searches, we dropped studies that did not fit the scope of this meta-analysis. Studies that focused on land remediation of coal mines or quarries were not included, since land remediation practices and goals substantially differ from grassland restorations. We did not include small-scale studies that took place within a larger agricultural or urban context, such as wildflower strips or pollinator gardens. Studies that focused on floral resources within the plant community and did not measure pollinator responses directly were not included. We also dropped studies using small-scale experiments (plots only a few meters in size) as pollinator responses are only expected to occur at larger spatial scales. For all papers we identified the means and standard deviations or standard errors of the treatments. When these were not reported, we contacted the authors for this data. These restrictions resulted in 25 studies we could use in our meta-analysis (listed in Supplementary Material S1).

Effect size calculations and data analyses

From the 25 studies, a total of 66 data points were obtained for analyses. To answer our two overarching questions, data points were first separated out into two large groups: (1) those that compared restorations to degraded sites, and (2) those that compared restorations to intact remnant sites. Some studies did both of these comparisons in their paper and so both data points were used. We further separated data points out by what metric of pollinator diversity they assessed: abundance or species richness. When studies compared restorations of different ages or at several time points, we chose those that were oldest and had been carried out for the longest time, similar to the methodology of previous studies e.g. (Fleeger et al. 2018; Copeland et al. 2018).

To answer our secondary questions, we divided study results into smaller groups as available. Taxon-specific responses were analyzed for bees and lepidopterans. Studies that measured bee responses as a whole did not separate out *A. mellifera* responses, so we were unable to evaluate these responses at finer taxonomic scales. Studies varied widely in exactly how grasslands were degraded, but most studies fit into one of two general categories: degradation by invasive species or degradation by agriculture. To evaluate effects of methods associated with restoration practices, we separated studies into two groups: those that implemented one method (e.g., reintroduction of grazing), and those that used multiple methods (e.g., grazing and burning). Finally, to assess the effect of restoration age (time since initiation

of efforts), we categorized studies as young (< 10 years) or old (> 10 years).

To measure effect sizes across studies in this meta-analysis, we calculated Hedge's *d* (Wasserman 1988) for each relevant study comparison. Hedge's *d* takes into account the study's sample size, variance and mean of each treatment. Effect sizes (*d*) along with the variation (*v*) for each study were calculated to assess restoration treatment significance using the R package metafor (Viechtbauer 2010). To compare effect size means across the different groups, we used randomized effect models carried out in R (Version 3.6.0), again, in metafor (Viechtbauer 2010). If a model was significant ($p < 0.05$), we deemed the restoration to be different from the compared site (degraded or remnant grasslands respectively).

Results

Both pollinator richness and abundance responded positively to grassland restorations when compared to degraded sites ("Full" analyses, Fig. 1; abundance: $d = 1.31$, $p = 0.015$; richness: $d = 0.894$, $p = 0.045$ respectively). When restorations

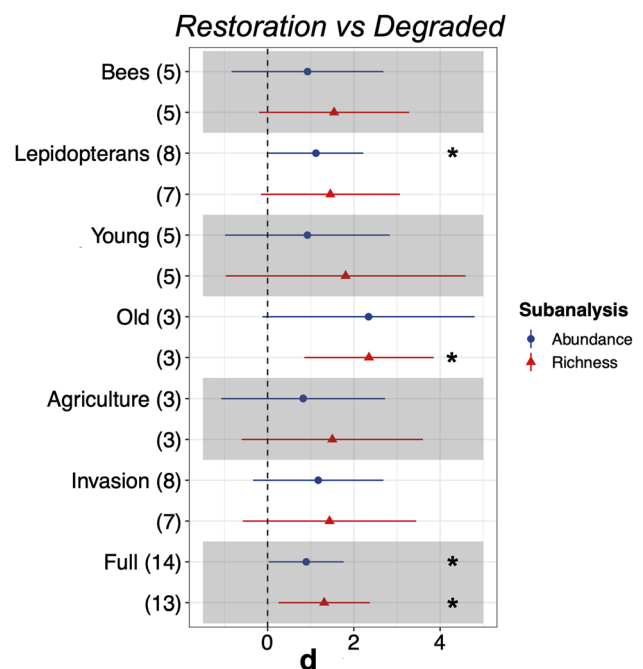


Fig. 1 Meta-analysis results for pollinator communities in grassland restorations compared to those in degraded grasslands. Effect size means and 95% CIs are depicted for each comparison. Abundance values are in blue circles and richness values are in red triangles. Values greater than zero (dashed line) indicate increases in pollinator measures compared to degraded sites. Sub-analysis groupings shown to the left (e.g. Bees) and sample sizes for each grouping are in parentheses. Statistically significant differences between restored and degraded sites at $p < 0.05$ level indicated with an asterisk (*)

were compared to remnant grasslands, pollinator communities were not significantly different, indicating that restorations were effective at restoring pollinators to target conditions (Fig. 2; abundance: $d = -0.366$, $p = 0.167$, richness: $d = -0.411$, $p = 0.253$).

Taxon responses

When the data points were separated by taxa, lepidopteran abundance in restorations was significantly improved compared to degraded sites (Fig. 1; $d = 1.120$, $p = 0.047$), though richness was not (Fig. 1; $d = 1.455$, $p = 0.07$). For bees, neither abundance nor richness were improved from degraded sites (Fig. 1; $d = 0.925$, $p = 0.301$; $d = 1.541$, $p = 0.082$).

When restorations were compared to remnants, bee abundance and richness did not differ (Fig. 2; abundance: $d = -0.811$, $p = 0.06$; richness: $d = -0.322$, $p = 0.194$). Lepidopterans showed similar responses (Fig. 2; abundance: $d = -0.176$, $p = 0.767$; richness: $d = -0.874$, $p = 0.326$).

Restoration age

There was a difference in response between young and old restorations (Fig. 1). Pollinator richness in older restorations

(> 10yrs) showed significant improvements compared to degraded sites ($d = 2.35$, $p = 0.002$). However, younger restorations did not show improvements from degraded sites (abundance: $d = 0.922$, $p = 0.345$; richness: $d = 1.81$, $p = 0.202$).

When comparing young and old restorations to remnant grasslands, both old (Fig. 2; abundance: $d = -0.33$, $p = 0.533$; richness: $d = 0.09$, $p = 0.86$) and young restorations (Fig. 2; abundance: $d = -0.32$, $p = 0.572$; richness: $d = -1.03$, $p = 0.189$) do not differ significantly from remnants.

Land degradation history

There were no significant differences in responses of pollinators to restoration of agricultural lands compared to lands with invasive species (Figs. 1, 2).

Restoration methods

We only had enough studies to look at these effects for lepidopterans ($n = 5$; Bee or other taxa studies $n = 0$). Lepidopteran abundance was influenced by restoration method, where restorations that implemented more than one method had a significant increase in abundance (Fig. 3; $d = 1.0722$, $p = 0.003$). However, richness was not affected by the number of methods used (Fig. 3; $d = 0.09$, $p = 0.815$).

Discussion

Our results show that grassland restorations substantially improve pollinator community diversity, though factors such as pollinator taxon, land degradation history, and restoration age can influence these outcomes. In fact, seven of the studies used in this meta-analysis saw pollinator communities (abundance/richness) *more than double* in the restoration sites when compared to degraded sites (Griffin et al. 2017; Maccherini et al. 2009; Rutgers-Kelly and Richards 2013; Alison et al. 2017; Lettow et al. 2018, 2007; Helbing et al. 2015). However, a few studies did not see any improvement from degraded sites. For example Emery and Doran (2013) removed an invasive forb from their grassland sites, which left floral resources much lower than in invaded sites.

Notably, grassland restoration efforts did not just improve pollinator communities when compared to degraded habitats, but actually restored communities to match those found in target remnant sites. However, the mean effect sizes from most studies were negative, indicating a general trend that pollinator communities still can fall short of restoration goals, possibility due to needing more time or lack of key plant species. Interestingly, it is possible for restoration efforts to actually result in pollinator communities that are

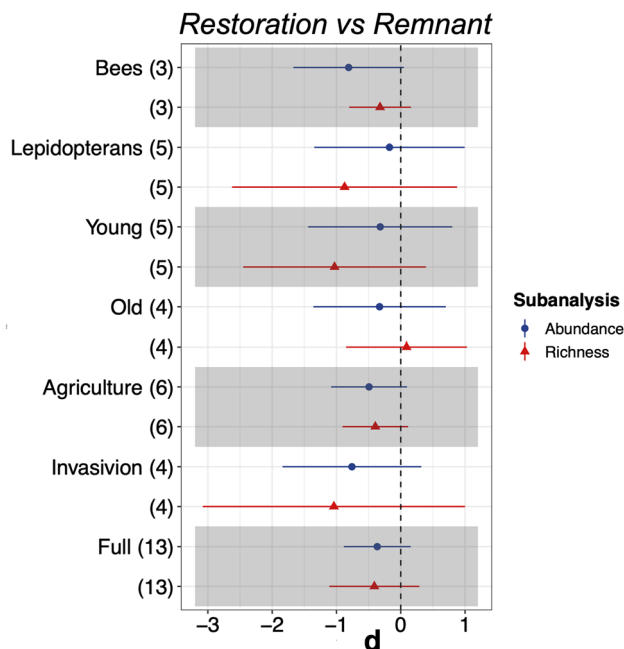
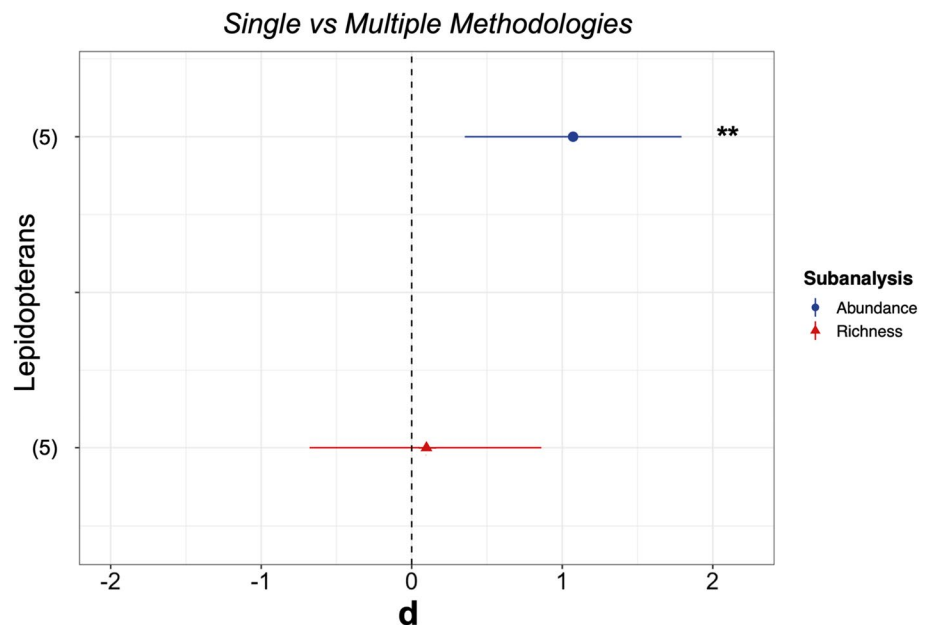


Fig. 2 Meta-analysis results for pollinator communities in grassland restorations compared to those in remnant grasslands. Effect size means and 95% CIs are depicted for each comparison. Abundance values are in blue circles and richness values are in red triangles. Values less than zero (dashed line) indicate decreases in pollinator measures compared to remnant sites. Sub-analysis groupings shown to the left (e.g. Bees) and samples sizes for each grouping are in parentheses. No comparisons were statistically significant at the $p < 0.05$ level

Fig. 3 Meta-analysis results for lepidopteran communities in grassland restorations using one or multiple restoration methods. Effect size means and 95% CIs are depicted for each comparison. Abundance values are in blue circles and richness values are in red triangles. Values greater than zero (dashed line) indicate increases in lepidopteran measures for restorations using multiple methods compared to single methods. Samples sizes for each grouping are in parentheses. Statistically significant differences between single and multiple methodologies at $p < 0.01$ level indicated with asterisks (**)



more diverse and abundant than remnant sites. For example, Ries et al. (2001) actually found increased lepidopteran abundance and richness values in restoration sites as compared to remnant sites in roadside tallgrass prairies. This may be due to depressed pollinator communities in isolated remnant prairies that are no longer interconnected at a regional scale.

Differences in pollinator taxa

Bees and lepidopterans represent the most important and common pollinator taxa in grasslands (Allen-Wardell et al. 1998; Archer et al. 2014). However, differences arise in how the two taxa respond to grassland restorations. For lepidopterans particularly, abundances rather than richness showed improvements in restorations. This may be because specific host plants of rarer lepidopteran species may not be included in a seed mix, or may be more difficult to support in restorations. Additionally, landscape factors may play a role, as butterfly specialists suffer the most from decreased habitat connectivity (Bruckmann et al. 2010). Finally it is possible that land managers may give extra attention to the success of particular flagship species, which may increase overall abundance of the focus species while having marginal effects on species richness.

The lack of clear improvements in bee communities may be due to the fact that *A. mellifera* were lumped together with native bees in most of these studies. *A. mellifera* does not rely on native grasslands for habitat, and tends to be a dominant species when it is present. Several studies have shown that *A. mellifera* can have negative effects on the wild bee community (Walther-Hellwig et al. 2006; Herbertsson

et al. 2016; Graystock et al. 2014; Lindstrom et al. 2016). Even if restoration efforts improve habitat for native bees, the presence of *A. mellifera* may swamp out any benefits.

Additionally, many bee species, both solitary and social, are ground nesting (USDA 2007). This nesting behavior may also drive the overall lack of response, as many restoration methods involve soil disturbances. However, single disturbance events are often beneficial as they can create open soil for ground dwelling species (Williams et al. 2010). This role of life history influencing where pollinators can nest and survive has already been documented in urban areas, where ground nesting bees are found in low numbers, but cavity nesters thrive (Fitch et al. 2019). As bees are the most important pollinators for most crop and wild species (Hanley et al. 2015; Allen-Wardell et al. 1998), more documentation and planning may be warranted when implementing grassland restorations.

Importance of time

The time subanalysis showed that it often takes 10 or more years for pollinator communities to be restored. This may be due to the fact that plant and soil communities often take several years to be fully restored. For example, soil nutrients or microbiota may be dramatically altered by agriculture or invasive species, and take decades to restore. McKee et al. (2019) found that in a chronosequence of prairie restorations, almost all measures of soil health, including organic matter, bulk density, carbon, and nitrogen increased with time since restoration. Similarly, Scott and Morgan (2012) found the plant community in old fields took between 20 and 40 years to reach levels similar to those of uncultivated

fields. During this time exotic species were slowly replaced with native cultivars and richness values were restored to remnant values. Additionally, it may take years for pollinators to disperse into a new restoration. For example, most Midwestern US prairie restorations are surrounded by agricultural land and therefore are habitat islands that can be difficult to recolonize, especially by those that are dispersal limited. (Samson and Knopf 1994). Öckinger et al. (2018) found that solitary bees were less likely to colonize a restoration than bumblebees and hoverflies and across all species, colonization was dependent on habitat connectivity.

Land degradation history

Degradation and land use history had minimal effects on the ability to restore pollinator communities. This is somewhat surprising as we expected agriculture to have particularly detrimental effects on pollinator communities due to regular tillage (Williams et al. 2010), and so be harder to restore. However, widespread usage of chemical herbicides to manage invasive plants could have strong deleterious non-target effects on pollinators (Motta et al. 2018; Balbuena et al. 2015), which may partially explain this lack of land use difference. Our findings are supported by at least one other study, which showed that agricultural history was not a good predictor of wild bees in pine savannah restorations (Odanaka et al. 2020).

Restoration methods

While we were only able to address this question for lepidopterans, we did find that restorations using multiple methods increased lepidopteran abundance, but not richness, compared to those that used only single restoration methods. We expected that multiple restoration methods might decrease pollinator diversity due to increased disturbance, which does not seem to be the case for lepidopterans. Instead, multiple methods may increase heterogeneity of habitat for pollinators (Glenn et al. 1992), increasing pollinator diversity. Only a handful of studies have addressed how management methods affect pollinator restoration though, and much more work is needed. Very little is known about how grazing, fire, tilling, and other restoration methods might interact to affect pollinator nesting and other behaviors. For example, Öckinger et al. (2018) found that pollinator richness was actually lower in restorations, possibly due to the reintroduction of grazers, which has been shown to have variable effects on pollinator communities (DeBano et al. 2016; Elwell, Griswold and Elle 2016; Hartnett, Hickman and Walter 1996). Several studies have shown that the net effects from grazers is highly dependent on the density, with low intensity grazing showing positive effects for pollinators (Sjodin et al. 2008; Albrecht et al. 2007).

Conclusions

Altogether, this meta-analysis shows that grassland restorations are important for pollinator conservation. These results are particularly relevant because they include large-scale studies that differ in land use history, location, and age but still show significant improvements in pollinator diversity and abundance. A clear difference arises between old and young restorations, indicating that practitioners should not expect a full recovery in the first few years.

If the funds are available, we cautiously suggest that employing two or more restoration methods may be useful, at least for lepidopterans. Restorations using more than one method saw significant increases in pollinator abundances compared to those using only one method. This finding is independent of the type of method used. For example, while we were not able to address whether fire was better than grazing or mowing, we can conclude that any combination of two methods was better than only one. However, there is a lack of research on how bees may respond to multiple restoration methods.

It is somewhat worrisome that bees showed weaker and more variable responses to restoration activities than lepidopterans. This may be because *A. mellifera* presence confounded responses of native bees in these studies. Native bees are vital pollinators for many grassland plant species and are often at the center of pollinator interaction networks (Hansen et al. 2018; Forup and Memmott 2005). Historically, little focus has been given to nesting resources for native bees in restorations, but these should be considered as much as the floral resources if restorations are to support a robust native bee community.

Finally, while grassland restorations clearly improve pollinator communities, the *d* values for restorations compared to remnant grasslands were consistently negative, indicating that pollinator communities still fall short from being fully restored. In a simple vote count of results that compared restorations to remnant communities, about half of all studies found that pollinator communities were not fully restored. This highlights the need to not just restore degraded grasslands, but also protect remnant grasslands for pollinator conservation. Ideally, land managers could strive to connect the remaining fragmented remnant grasslands with corridors of restored grasslands or increase the size of remnant patches with adjacent restorations.

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Author contributions AS, SE conceived the study, AS performed data collection and analyzed the data, AS, SE wrote and edited the manuscript.

Compliance with ethical standards

Conflict of interest The authors declare no conflicts of interest for this study.

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